

LandWeb v3: Overview

Historic Natural Range of Variation modelling in the western boreal — and how to read this report suite

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2026-07-03

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Warning

Draft report: pending human review

Portions of this document were drafted with the help of generative AI (large language models) and have not yet been fully reviewed by the co-authors. Content may be incomplete, inaccurate, or misattributed. This notice will be removed once a complete review has been carried out.

1 What LandWeb is

LandWeb is a data-driven modelling configuration for simulating the **Historic (Natural) Range of Variation (H/NRV)** of forest condition across most of the western Canadian boreal forest (~3 million km²). Rather than a single model, it couples two well-established simulation models inside the SpaDES platform (Chubaty and McIntire 2025) — a workflow embodiment of the PERFICT approach to predictive ecology (McIntire et al. 2022; Barros et al. 2023):

- **vegetation dynamics** — an implementation of the LANDIS-II Biomass Succession model (Scheller and Mladenoff 2004; Scheller et al. 2007), simulating growth, mortality, dispersal, and post-disturbance regeneration of tree-species cohorts on each pixel; and
- **wildfire** — an implementation of the LandMine landscape-disturbance model (Andison 1996; Andison 1998), igniting and spreading fires at rates set by a long-term historic fire-cycle (LTHFC) map.

Run over many centuries and many stochastic replicates, the simulation produces a *distribution* of plausible pre-industrial landscape states — the natural range of variation — against which the current landscape can be compared. The pre-industrial fire regime that anchors the fire model comes from a long-term historic fire-cycle synthesis for the western boreal (Andison 2019).

2 The v3 update at a glance

v3 is primarily a **rebuild of the modelling infrastructure and a refresh of the input data**, not a change of ecological content. The main changes (each documented in its own report, below):

- **Inputs.** Tree-species / biomass / age initial conditions move from the Beaudoin *k*NN 2001 maps (Beaudoin et al. 2014) to SCANFI 2020 (Guindon et al. 2026) (report 01); the fire-cycle map is updated to **v10**, incorporating the June-2026 Northwest Alberta work (Peterson et al. 2026) (report 02).

- **Orchestration.** The manual SpaDES script chain is replaced by a **targets pipeline** that owns caching and dispatch, with the SpaDES modules retained as the simulation engine (report 03).
- **Scope.** Simulations are run and reported **per forest-management area (FMA)** rather than as one monolithic run.
- **Post-processing.** Three v2 summary modules are consolidated into the new NRV_summary module plus the nrvtools package (report 03).

3 Modelling workflow

The **targets** pipeline runs the data-preparation stage once per study area (caching the study-area-independent species trait factorial), then crosses the prepared data with a set of stochastic replicates for the main simulation, and reduces the replicates into the NRV summaries. The generic SpaDES-to-targets glue lives in the owned SpaDES.targets package; heavy stages are dispatched across a small compute cluster via **crew.ssh** (report 03).

4 Study areas

v3 treats the **forest-management area (FMA)** as the unit of simulation and reporting: each area is parameterized, simulated, and summarized on its own, rather than as part of a single continental run (the v2 LandWeb_full run is dropped). This keeps per-area updates from invalidating unrelated results and keeps peak memory bounded. The FMAs span the western boreal across British Columbia, Alberta, Saskatchewan, Manitoba, and the Northwest Territories; the historical per-FMA client deliverables in **reports/2020/** indicate the scope. The v3 pipeline was first validated end-to-end on the **Spray Lake** study area (SW Alberta).

5 This report suite

The v3 documentation is organized as a numbered suite (this overview is report 00), plus the separate user manual. Reports are rendered from the pipeline so their figures and numbers stay consistent with the model.

Table 1: The LandWeb v3 report suite.

Report	Contents
00	Overview and rationale (this document); modelling workflow; report index.
01	Study-area vegetation: SCANFI composition, and the kNN (v2) to SCANFI (v3) initial-condition change.
02	Fire-model inputs: the LandMine LTHFC v10 (2026 NW Alberta) update vs the previous v8c / Spray Lake layers.

03	Code and module changes v2 to v3: per-module/package history; SpaDES framework breakage and the targets switch.
04	Output differences v2 vs v3: how the simulated NRV and burn summaries change (planned).
05	Per-study-area summary: the templated NRV report rendered per FMA — the client-facing deliverable (planned).
06	Provenance: the input/output manifest and reproducibility standards (planned).
07	Best-practices alignment vs forest-landscape-model calibration guidance (planned, optional).
Manual	User guide / manual (migrating from bookdown to a Quarto book), tracked by issue #128.

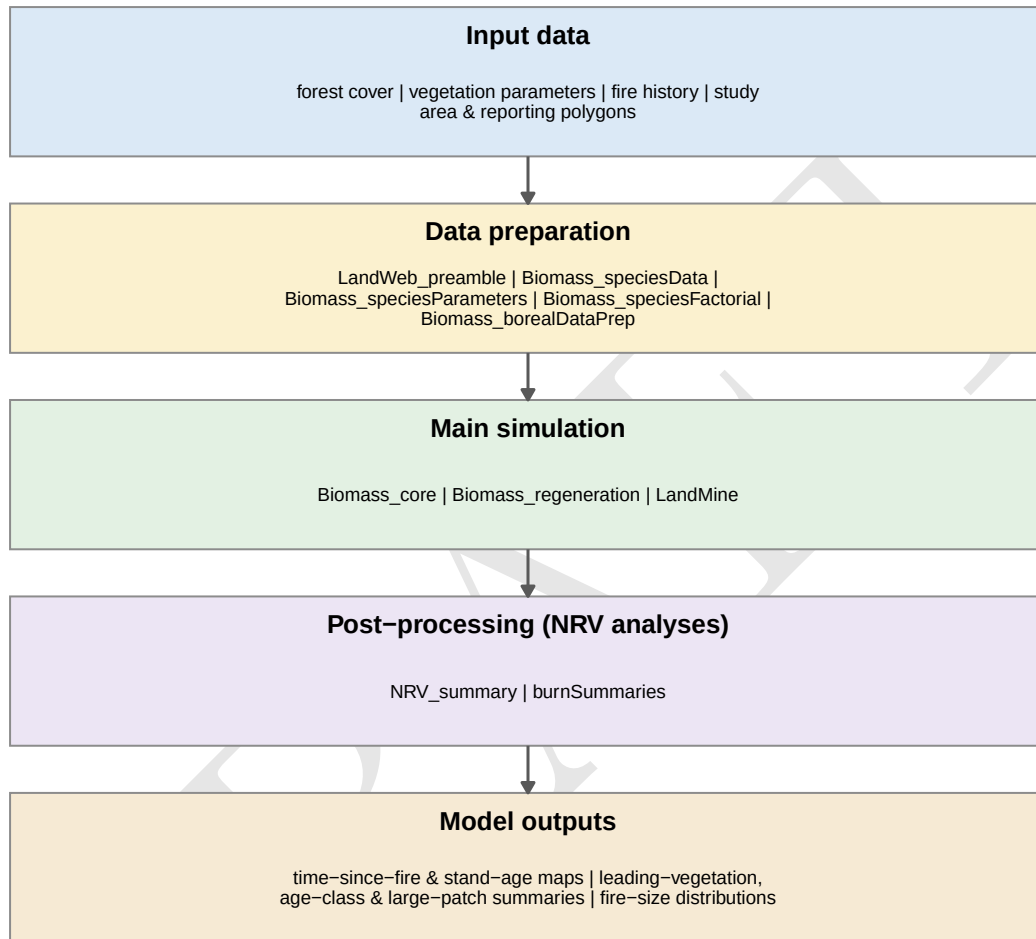


Figure 1: The LandWeb v3 modelling workflow: input data feed a data-preparation stage, a main simulation, and a post-processing (NRV analyses) stage that produces the summary outputs. In v3 the post-processing modules NRV_summary and burnSummaries replace the v2 trio timeSinceFire / LandWeb_output / LandWeb_summary.

References

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DRAFT

Provenance

- **Source:** this overview is narrative; the workflow figure (Figure 1) is a fixed schematic built in `ggplot2` from the current `v3` module set (cf. the DiagrammeR source in `R/model_workflow.R`).
- **Method:** rendered by the gated `landweb.extended_analyses` factory as a `tar_quarto()` target on the control node; no pipeline data dependencies.

Appendix: Build provenance

This appendix pins the rendered PDF to the code, environment, and revisions that produced it, so a result can be reproduced or a difference attributed to code vs. inputs vs. randomness. It is generated at render time from the `workflowtools info_*`() family.

Build identity. LandWeb repository, branch `targets`, commit `ad683998`, rendered 2026-07-03 10:18 MDT with R version 4.6.1 (2026-06-24) on Ubuntu 26.04 LTS (GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1).

Table 2: SpaDES modules and owned packages at their pipeline-pinned revisions when this report was rendered (`workflowtools::info_submodules()`).

Component	Kind	Pinned SHA
Biomass_borealDataPrep	module	f570094ea9
Biomass_core	module	32a3790870
Biomass_regeneration	module	b739da2d32
Biomass_speciesData	module	101ad29571
Biomass_speciesFactorial	module	01d4559d15
Biomass_speciesParameters	module	1e381a7881
HSI_Caribou_MB	module	1180d61743
LandMine	module	d7ccb083e1
LandWeb_output	module	af78cee648
LandWeb_preamble	module	65dc365f37
LandWeb_summary	module	99e229e3ac
NRV_summary	module	ca7b444fa2
burnSummaries	module	add6cc0bb4
timeSinceFire	module	30d242ea7b
LandWebUtils	package	cf29ea06e4
SpaDES.targets	package	b1f68a5467
crew.ssh	package	79c368f0fa
nrvttools	package	f9665fd7b0
spatialutils	package	a891aef9d5
workflowtools	package	561def4389

The full `sessionInfo()` and the exact `renv.lock` package pins accompany the run; input datasets are recorded in `_input_manifest.json` (report 06).